
Microprocessor Systems and Interfacing

EEE 342

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Lecture # 10

Week 7 Lecture 1

Microprocessor 8088

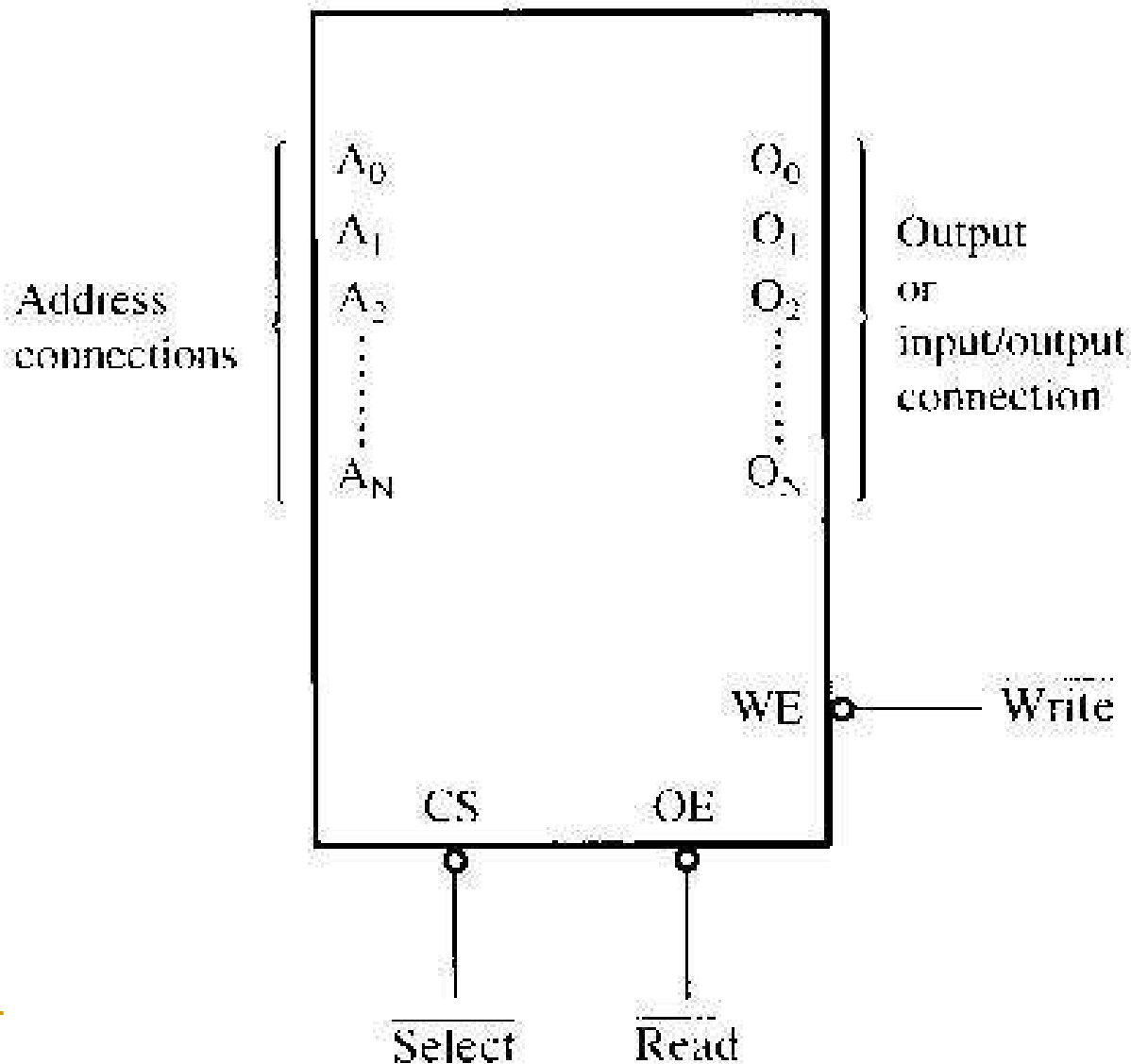
■ Outline

- Memory interfacing with 8088 using 74LS138 Decoder

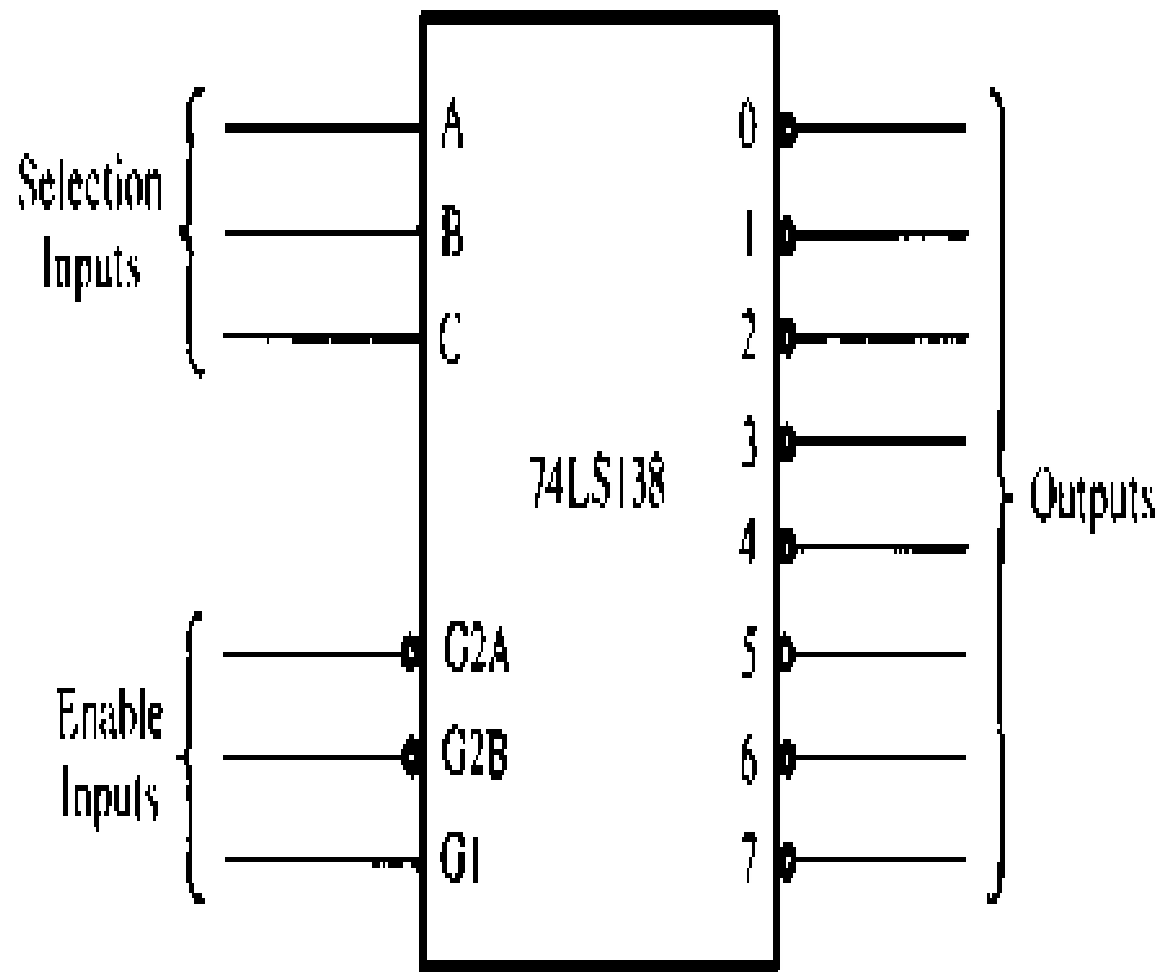
Memories

- Some Important numbers to consider
 - ❑ 1K = 1024 Location → 10 Address Pins, 2^{10}
 - ❑ 2K = 2048 Location → 11 Address Pins 2^{11}
 - ❑ 4K = 4096 Location → 12 Address Pins 2^{12}
 - ❑ 1M = 1 Mega Location → 20 Address Pins
 - ❑ 400H represents 1K-byte of Memory
 - ❑ 1000H represents 4K
 - ❑ 2000H represents 8K

Pin configuration of Memory Chips



The 3-to-8 Line Decoder (74LS138)



Inputs						Outputs							
Enable			Select										
$\overline{G2A}$	$\overline{G2B}$	$G1$	C	B	A	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	7
1	X	X	X	X	X	1	1	1	1	1	1	1	1
X	1	X	X	X	X	1	1	1	1	1	1	1	1
X	X	0	X	X	X	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	0

The Line Decoder
(74LS138)

NAND Gate Decoder

- Interface 16KB SRAM with 8088 using NAND Decoder. Available chip size is 4KB. Memory map starts at address 8E000h

Using Line Decoder

- Interface 64KB EPROM with 8088 using LINE Decoder. Available chip size is 16KB. Memory map starts at address 40000h